

DITING: A Weak Degradation Listener for Battery Lifetime Early Prediction

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Abstract

Battery lifetime early prediction is crucial for safety assessment and decision planning, yet early-stage degradation signals are extremely weak and difficult to distinguish from stochastic noise. Existing methods primarily rely on denoising or signal decomposition, which may lose critical degradation cues. In nature, most organisms exhibit the binaural effect, exploiting discrepancies between left and right auditory inputs to enhance perceptual reliability. Inspired by this, we propose *DITING*, a *weak degradation listener* for battery lifetime early prediction. We first employ optimal-transport-based selective matching to extract a robust health template from initial cycles, and further design a tri-coupled degradation manifestation mechanism to distinguish degradation signals from noise. By exploiting the randomness of noise, matched responses under symmetric coupling suppress stochastic fluctuations, while degradation-driven cumulative deviations propagate through the coupling process to form stable bilateral discrepancies, thereby amplifying weak early-stage cues for lifetime prediction. Experiments on various datasets demonstrate that DITING achieves state-of-the-art performance and provides more reliable early support for full-lifecycle battery management.

1. Introduction

Accurate early prediction of battery lifetime is critical for health management in new energy vehicles, energy storage systems, and intelligent equipment (Thelen et al., 2024). As the cornerstone of electric mobility and renewable grid storage, batteries typically exhibit stochastic and nonlinear

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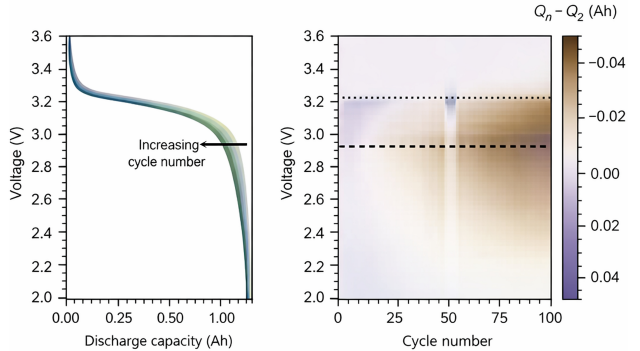


Figure 1. Early-cycle voltage and capacity variations. Due to noise, voltage curves and inter-cycle capacity differences in early cycles do not exhibit a monotonic degradation evolution.

degradation trajectories. Early studies primarily relied on physics-based models that describe degradation through simplified electrochemical or empirical dynamics, which often require strong assumptions and extensive domain knowledge (Dong & Wei, 2021; O’Kane et al., 2022). Subsequent statistical approaches formulated lifetime prediction as a stochastic process with handcrafted features and parametric distributions, offering interpretability but limited flexibility (Zhang et al., 2023b). With the increasing availability of large-scale observed data, data-driven methods have become dominant. Deep learning approaches typically employ time series modeling architectures such as recurrent neural networks, convolutional neural networks, and Transformers as backbones, utilizing meticulously designed mechanisms to learn latent cross-cycle and cross-channel correlations from cycling data (Tang et al., 2022; Han et al., 2023; Yao et al., 2025; Zhang et al., 2025). Recently, BatteryML (Zhang et al., 2024a) and BatteryLife (Tan et al., 2025) have introduced unified evaluation protocols and benchmark suites, improving the reproducibility and comparability of method comparisons. Despite these advancements, inferring battery lifetime from early-cycle data remains challenging, as weak degradation signals are frequently confounded with noise, as shown in Figure 1 (Attia et al., 2021).

In real-world scenarios, recorded degradation trajectories are inevitably corrupted by stochastic noise caused by vary-

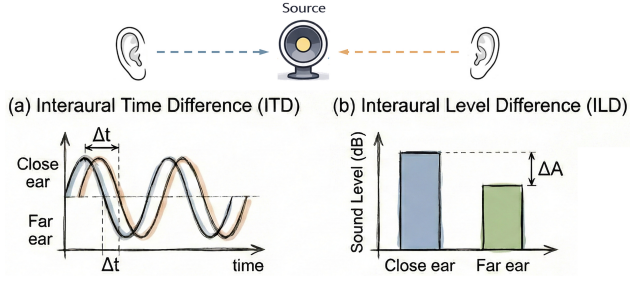


Figure 2. Binaural effect in biological systems. Binaural hearing exploits discrepancies between left and right auditory inputs, with interaural time difference and interaural level difference providing complementary cues to enhance perceptual reliability.

ing operating conditions and unstable loads. To mitigate such interference, existing methods have developed three broad categories of strategies. First, classical signal processing methods apply filters or wavelet-based denoising to suppress high-frequency perturbations (Halidou et al., 2023; Frusque & Fink, 2024). Beyond the above approaches, decomposition-based architectures explicitly disentangle temporal variations into trend, seasonal, and residual components, improving robustness under long-horizon dynamics (Cao et al., 2026; Yu et al., 2024; Cao et al., 2024). Furthermore, diffusion-based generative models have been adapted to time series forecasting, where iterative denoising provides a distributional modeling perspective (Shen et al., 2024; Liu et al., 2025; Pei et al., 2025). Relatedly, flow matching objectives have also been explored for modeling continuous-time trajectory dynamics (Zhang et al., 2024b). Although these designs are effective when signal and noise are relatively separable, they typically rely on a clear boundary between them. At the early degradation stage, the aging signature is intrinsically weak and entangled with measurement noise. This coupling undermines standard pipelines in practical settings, since aggressive suppression can misidentify weak degradation cues as noise.

Intriguingly, biological auditory systems can perceive weak acoustic cues in complex environments by comparing bilateral sensory inputs. This phenomenon is commonly referred to as the binaural effect (Zhang et al., 2023a). It supports robust auditory scene construction through central integration of bilateral inputs. As illustrated in Figure 2, interaural time difference and interaural level difference provide complementary spatial cues for sound localization (Carlini et al., 2024). Crucially, bilateral comparison can reduce the influence of channel-specific perturbations, thus improving perceptual reliability.

Motivated by this insight, we reformulate early battery degradation identification from single observation modeling

to discrepancy-based comparison. Battery aging is driven by irreversible electrochemical reactions, and early capacity fade is often dominated by the steady growth of the solid electrolyte interphase, yielding a directional and structurally consistent trajectory over the life cycle (Li et al., 2024; Chen et al., 2024). In contrast, stochastic noise typically lacks a consistent direction and is unlikely to induce stable comparative patterns. Our objective is therefore to expose genuine weak degradation trends from stochastic perturbations, rather than merely suppressing noise. Appendix A provides the full biological motivation.

In this paper, we propose *DITING*, a *weak degradation listener* named after a mythical creature in Chinese folklore that is known for distinguishing truth from chaos. To identify weak degradation signals from early cycles, DITING first employs Sinkhorn optimal transport and weighted medoid fusion to construct a robust health prototype. This procedure aligns the distributional geometry of early cycles to form a structural reference and derive degradation residuals. We then introduce a tri-coupled degradation manifestation mechanism, where the middle branch receives the degradation residuals and the left and right branches are driven by anti-phase sinusoidal excitations. Owing to the randomness of noise, the symmetric opposing updates tend to attenuate fluctuations across side branches. In contrast, genuine degradation is directionally consistent and cannot be neutralized by the left and right symmetric branches, which makes it manifest as a stable discrepancy between the left and right outputs. The resulting manifested representation is finally used for battery lifetime prediction. By quantifying the bilateral output discrepancy, DITING extracts weak degradation features from noisy early-cycle backgrounds and enables accurate prediction at the early stage of battery usage. Our main contributions are summarized as follows:

- We propose DITING, a bio-inspired weak degradation listener that establishes a discrepancy-aware paradigm for early battery lifetime prediction by characterizing the contrast between stochastic noise and consistent degradation dynamics.
- We construct a health prototype using Sinkhorn optimal transport and weighted medoid fusion, providing a structural reference for deriving degradation residuals from early cycles.
- We design a tri-coupled degradation manifestation mechanism with bilateral anti-phase responses, attenuating stochastic noise and manifesting genuine degradation as a stable left-right discrepancy.
- Experiments on various battery datasets demonstrate that DITING achieves high accuracy for battery lifetime early prediction even under noise interference.

2. Related Work

2.1. Battery Lifetime Early Prediction

Accurate prediction of battery lifetime facilitates reliability assessment and system-level optimization. Advancements in prognostic methods have evolved from physics-based empirical models (O’Kane et al., 2022; Dong & Wei, 2021) to deep learning architectures including recurrent neural networks, convolutional neural networks, and Transformers (Zhang et al., 2024a; Tan et al., 2025; Zhang et al., 2025). Deep models capture latent cross-cycle correlations and benefit from large-scale benchmarking efforts that standardize evaluation protocols (Zhang et al., 2024a; Tan et al., 2025). Meanwhile, battery foundation models leverage pre-training on diverse electrochemical data to support data-driven lifetime prediction across operating conditions (Li et al., 2025). Additionally, generative synthesis has been explored to augment training distributions, where diffusion processes enrich early-cycle observations (Eivazi et al., 2024). Despite these advancements, inferring lifetime from early cycles remains challenging as weak degradation signals are entangled with stochastic noise, often leading to overfitting or signal suppression in standard architectures.

2.2. Stochastic Noise Mitigation in Time Series

Mitigating stochastic interference is fundamental for robust time series modeling. Classical strategies typically employ frequency-domain filtering or wavelet-based denoising to smooth raw observations (Frusque & Fink, 2024; Ouyang et al., 2023). Moving beyond simple smoothing, decomposition-based architectures separate temporal variations into trend, seasonal, and residual components to isolate key dynamics from noise (Wu et al., 2021; Woo et al., 2023; Ge et al., 2025). Furthermore, motivated by advances in generative modeling, diffusion-based approaches have been explored for time series forecasting and denoising through iterative refinement (Shen & Kwok, 2023; Shen et al., 2024; Wang & Ventre, 2024). Relatedly, flow matching provides a continuous-time alternative by learning vector fields that transport noise to data distributions (Lipman et al., 2023; Kollovieh et al., 2025). These methods are effective when informative patterns remain distinguishable, but they often rely on a relatively separable boundary between signal and noise. In this work, we introduce a discrepancy-aware mechanism that leverages the disparity between consistent degradation dynamics and stochastic perturbations to make weak degradation signals more identifiable at the early stage.

3. Task Definition

This section defines the task of early battery lifetime prediction. Standard degradation tests are conducted under preset charge and discharge protocols and controlled ther-

mal environments. For the i -th cycle, the voltage $V(t)$ and current $I(t)$ signals are recorded over the duration τ_i . The cycle discharge capacity Q_i is quantified by integrating the current magnitude over the discharge period, formulated as:

$$Q_i = \int_0^{\tau_i} |I(t)| dt \quad (1)$$

The state of health (SOH) at cycle i , a standard degradation indicator, is defined as the ratio of the current discharge capacity to the initial nominal capacity Q_0 :

$$\text{SOH}_i = \frac{Q_i}{Q_0} \quad (2)$$

Following the standard convention in prognostics, the battery cycle life Y is determined as the first cycle index at which SOH drops below a threshold of 80%, marking the end-of-life point (Severson et al., 2019; Attia et al., 2020). Formally, we represent the monitoring data for the i -th cycle as a multivariate time series matrix $\mathbf{X}_i \in \mathbb{R}^{S \times C}$, where S denotes the number of sampling points within the cycle and C denotes the sensor variable dimensions, including voltage, current, and capacity. The early degradation trajectory is thus denoted as $\mathcal{X}_{1:T} = [\mathbf{X}_1, \mathbf{X}_2, \dots, \mathbf{X}_T]$, where T denotes the cutoff index of the early observation window, e.g., $T = 100$ (Severson et al., 2019; Zhang et al., 2025). The objective of early battery lifetime prediction is to learn a nonlinear mapping function $\mathcal{F} : \mathcal{X}_{1:T} \rightarrow Y$, which predicts the total useful life Y utilizing exclusively the early cycling data $\mathcal{X}_{1:T}$. This task requires the model to establish the subtle relation between early degradation patterns in $\mathcal{X}_{1:T}$ and the long-term lifetime Y under stochastic noise and heterogeneous aging conditions.

4. DITING: A Weak Degradation Listener

We propose DITING, a discrepancy-aware framework for early battery lifetime prediction under stochastic noise. As shown in Figure 3, DITING consists of three key components: (1) **Hierarchical Embedding**, which maps raw multivariate cycles into a latent representation space, (2) **Health Prototype Construction**, which uses Sinkhorn optimal transport and weighted medoid fusion to build a geometrically aligned structural reference, and (3) **Tri-Coupled Degradation Manifestation**, which models degradation residuals through bilateral anti-phase responses. Acting as a weak degradation listener, DITING transforms early-cycle observations into health-referenced residuals, attenuates stochastic noise through opposing side-branch responses, and manifests consistent degradation as a stable left-right discrepancy. The resulting discrepancy representation is finally used for battery lifetime prediction.

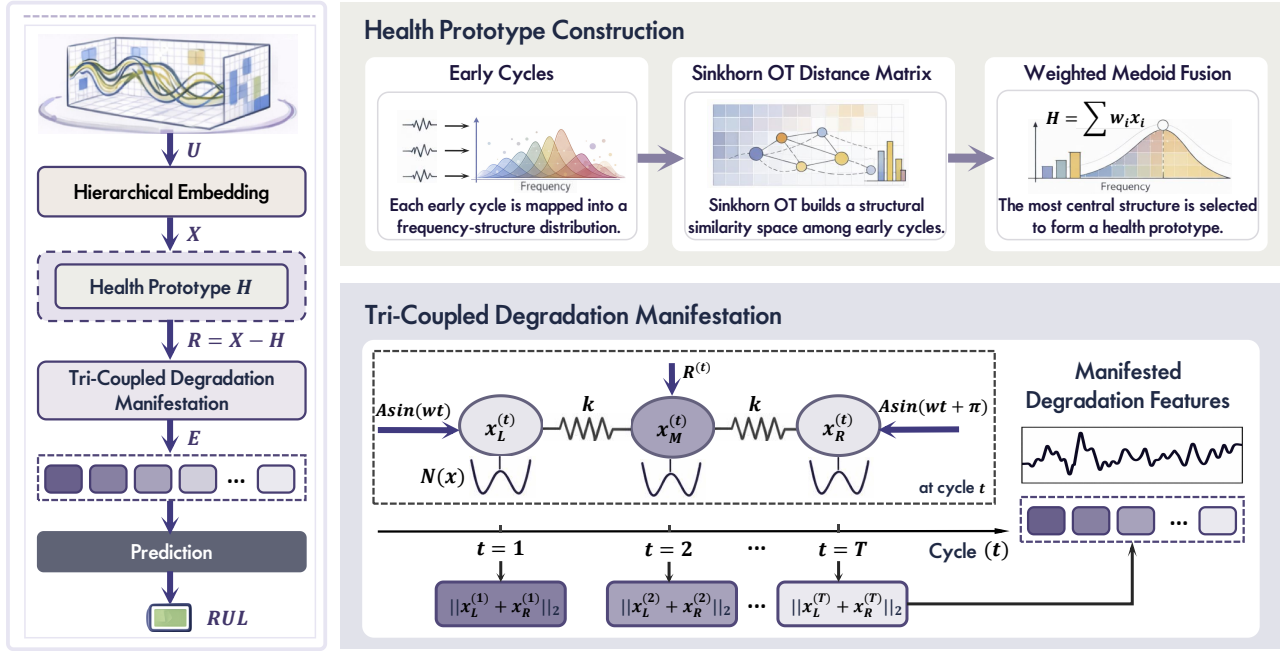


Figure 3. Overall architecture of DITING.

4.1. Hierarchical Embedding

Given the early-cycle observation data $\mathbf{U} \in \mathbb{R}^{T \times S \times C}$, where T denotes the number of observed cycles, S represents the number of sampling points per cycle, and C corresponds to the sensor variables. We first project the raw observations into a latent feature space via a linear transformation Φ . To preserve the complex temporal and semantic dependencies inherent in battery data without relying on recurrent structures, we introduce a hierarchical embedding mechanism. Specifically, we map each cycle into a D -dimensional vector with $D = SC$ and obtain a cycle representation sequence $\mathbf{X} \in \mathbb{R}^{T \times D}$, which serves as the input to the health prototype construction module in Figure 3. To encode the multi-scale structure of cycling data, we inject positional information along three dimensions, corresponding to the cycle order, the within-cycle sampling order, and the sensor variable type. The resulting embedding is formulated as:

$$\mathbf{X} = \Phi(\mathbf{U}) + \mathbf{P}^{(1)} + \mathbf{P}^{(2)} + \mathbf{P}^{(3)}. \quad (3)$$

Here, $\mathbf{P}^{(1)}$ applies rotary positional embeddings to model the irreversible temporal order across cycles (Su et al., 2024). $\mathbf{P}^{(2)}$ encodes the within-cycle sampling order using sinusoidal functions (Vaswani et al., 2017). $\mathbf{P}^{(3)}$ uses learnable embeddings to differentiate the semantic roles of sensor variable types, following variate tokenization in modern time series Transformers (Liu et al., 2024).

4.2. Health Prototype Construction

To derive genuine degradation patterns from the embedded sequence $\mathbf{X} = [\mathbf{x}_1, \dots, \mathbf{x}_T]$, we construct a robust reference baseline, termed the health prototype $\mathbf{H}$, to reduce the influence of stochastic measurement noise and capture the representative structural geometry of healthy cycles. This design is motivated by the empirical observation that the earliest cycles, even the first one or two cycles, can be strongly perturbed by transient fluctuations and measurement noise, and therefore may not reliably represent a stable healthy baseline. Rather than relying on simple arithmetic averaging, which is sensitive to outliers and phase shifts, we construct $\mathbf{H}$ based on the Wasserstein medoid of the early-cycle distributions. This construction leverages optimal transport geometry (Villani, 2008; Cuturi, 2013) to identify the most central cycle among the first K early cycles, which minimizes the aggregate transport cost to this initial early-cycle ensemble.

The construction follows a spectral optimal transport pipeline. First, each embedded cycle $\mathbf{x}_i$ is transformed into a probability measure μ_i via power spectral density estimation, which emphasizes structural periodicity and reduces the influence of transient fluctuations. Subsequently, the pairwise transport cost matrix $\mathbf{D} \in \mathbb{R}^{K \times K}$ is computed using the Sinkhorn algorithm (Cuturi, 2013). The elements D_{ij} quantify the entropy-regularized Wasserstein distance between cycles i and j , obtained by minimizing the follow-

ing transport objective. Intuitively, this distance measures the minimum cost of transforming the spectral distribution of one cycle into that of another. A smaller value of D_{ij} indicates higher structural similarity between the two cycles:

$$D_{ij} = \min_{\Pi \in \mathcal{U}(\mu_i, \mu_j)} \langle \Pi, \mathbf{M} \rangle - \varepsilon H(\Pi). \quad (4)$$

Here, $\mathcal{U}(\mu_i, \mu_j)$ denotes the transport polytope, the ground cost $\mathbf{M}$ incorporates both frequency shifts and cross-channel discrepancies, and the entropic term $\varepsilon H(\Pi)$ encourages smoothness in the transport plan.

Based on this transport geometry, the health prototype is constructed through a centrality-weighted fusion mechanism. The geometric medoid index is identified as $i^* = \arg \min_i \sum_j D_{ij}$. The prototype is then computed as a weighted sum of embeddings, with weights that decay exponentially with the transport distance to the medoid:

$$\mathbf{H} = \sum_{i=1}^K \frac{\exp(-D_{i,i^*}/\tau)}{\mathcal{Z}} \mathbf{x}_i, \quad (5)$$

where τ is a temperature parameter and $\mathcal{Z}$ is the normalization factor. This process forms a stable structural center for the early healthy state. The degradation residuals are then obtained by subtracting this prototype from each embedded cycle, $\mathbf{R} = \mathbf{X} - \mathbf{H}$.

4.3. Tri-Coupled Degradation Manifestation

We propose the Tri-Coupled Degradation Manifestation mechanism to reveal early weak degradation cues under noisy interference. It maps the input into left-right mirrored responses and couples them through a centrally anti-symmetric state, so that stochastic perturbations do not introduce a persistent directional bias, while key changes caused by degradation form stable bilateral discrepancies. To characterize this property, we give Theorem 4.1.

Theorem 4.1. *Consider a symmetric coupled system with anti-phase driving and centrally anti-symmetric response. Given an input $\mathbf{R}(t) = \mathbf{d}(t) + \mathbf{n}(t)$, where $\mathbf{n}(t)$ denotes zero-mean stochastic noise and $\mathbf{d}(t)$ denotes the degradation component, the signed bilateral offset induced by $\mathbf{n}(t)$ has zero expectation at symmetric equilibrium, i.e., $\mathbb{E}[\mathbf{Z}_{\text{noise}}] = 0$. Thus, stochastic noise does not generate a persistent directional drift in the bilateral discrepancy. In contrast, $\mathbf{d}(t)$ breaks the dynamic equilibrium and induces a stable nonzero signed bilateral offset in expectation, i.e., $|\mathbb{E}[\mathbf{Z}_{\text{signal}}]| > 0$, which is then accumulated as positive discrepancy energy. The proof is provided in Appendix B.*

Based on this theorem, we construct a dynamic environment where stochastic noise tends to cancel through mirrored coupling, while degradation cues are transformed into stable discrepancy representations for lifetime prediction. This

mechanism is implemented in two parts. The three-coupling linkage builds the structural basis for mirrored degradation manifestation, and the dynamic update process governs the state evolution that produces discrepancy energy.

Three-coupling linkage mechanism To reveal weak degradation from the health-prototype-corrected representation, we construct a dynamic field composed of three interacting branches: the left branch $\mathbf{x}_L$, the middle branch $\mathbf{x}_M$, and the right branch $\mathbf{x}_R$. The left and right branches form mirrored responses, while the middle branch serves as a central coupling state that coordinates their anti-symmetric evolution. To enhance the robustness of this dynamical process against transient stochastic interference, we introduce a second-order damped driven harmonic oscillator model (Wiggins, 2003). This second-order formulation requires accumulated driving energy to induce substantial state evolution, thereby preventing the system from being directly dominated by transient perturbations. Directionally unstable random noise is unlikely to sustain state deviations, and their effects gradually weaken under the damping effect. In contrast, persistent changes induced by degradation continuously inject driving energy and progressively form distinguishable dynamic discrepancies.

The dynamic field is described by three coupled second-order equations, where the left and right branches are driven in anti-phase and the middle branch receives the degradation representation $\mathcal{R}(t)$:

$$\ddot{\mathbf{x}}_L^{(t)} + \gamma \dot{\mathbf{x}}_L^{(t)} + \mathcal{N}_r(\mathbf{x}_L^{(t)}) + \mathbf{\Gamma}_L^{(t)} = A \sin(\omega t). \quad (6)$$

$$\ddot{\mathbf{x}}_M^{(t)} + \gamma \dot{\mathbf{x}}_M^{(t)} + \mathcal{N}_l(\mathbf{x}_M^{(t)}) + \mathbf{\Gamma}_M^{(t)} = C_M \mathcal{R}(t). \quad (7)$$

$$\ddot{\mathbf{x}}_R^{(t)} + \gamma \dot{\mathbf{x}}_R^{(t)} + \mathcal{N}_r(\mathbf{x}_R^{(t)}) + \mathbf{\Gamma}_R^{(t)} = A \sin(\omega t + \pi). \quad (8)$$

where the coupling terms $\mathbf{\Gamma}_L^{(t)}$, $\mathbf{\Gamma}_M^{(t)}$, and $\mathbf{\Gamma}_R^{(t)}$ are given by $\mathbf{\Gamma}_L^{(t)} = \kappa(\mathbf{x}_L^{(t)} - \mathbf{x}_M^{(t)})$, $\mathbf{\Gamma}_M^{(t)} = \kappa(2\mathbf{x}_M^{(t)} - \mathbf{x}_L^{(t)} - \mathbf{x}_R^{(t)})$, and $\mathbf{\Gamma}_R^{(t)} = \kappa(\mathbf{x}_R^{(t)} - \mathbf{x}_M^{(t)})$. Here, $\ddot{\mathbf{x}}$ denotes state acceleration, and $\dot{\mathbf{x}}$ denotes state velocity. A , C_M , κ , γ , and ω are hyperparameters representing the driving amplitude, degradation-representation scaling factor, coupling strength, damping coefficient, and driving frequency, respectively. The anti-phase driving terms $A \sin(\omega t)$ and $A \sin(\omega t + \pi)$ provide mirrored responses for the left and right branches, while these coupling terms constrain the linkage among the three branches through the middle state and stabilize their joint evolution. To characterize nonlinear degradation evolution, we introduce the nonlinear response function $\mathcal{N}(\cdot)$. The left and right branches share the same nonlinear mapping $\mathcal{N}_r$ to maintain the mirrored symmetric structure, whereas the middle branch adopts an independent $\mathcal{N}_l$ to adapt to the degradation representation input.

Crucially, the left and right branches are driven by forces with a phase difference of π . The degradation representation

$\mathcal{R}(t)$ is injected into the middle branch and transmitted to the two sides through the coupling term κ . Under the zero-mean noise condition used in the theorem, stochastic perturbations have unstable directions and are difficult to provide persistent directional energy. Their effects tend to cancel within the mirrored geometry, allowing the system to maintain the approximate anti-symmetric relation $\mathbf{x}_L \approx -\mathbf{x}_R$. In contrast, the degradation component $\mathbf{d}(t)$ induces a persistent shift of the middle state. This shift introduces an asymmetric bias into the coupled field, breaks the dynamic equilibrium between the anti-phase branches, and forms a measurable discrepancy between the bilateral responses.

Standard neural approximations do not necessarily preserve parity symmetry, and thus may produce unbalanced responses to the positive and negative phases of the driving force. Such unbalanced responses weaken the cancellation effect of stochastic perturbations in the mirrored structure. To address this, we propose a symmetrized non-linear construction $\mathcal{N}(\mathbf{x}) = \phi(\mathbf{x}) - \phi(-\mathbf{x})$, where ϕ consists of a two-layer linear network with Tanh activation. This structure satisfies strict central odd symmetry $\mathcal{N}(-\mathbf{x}) = -\mathcal{N}(\mathbf{x})$, allowing perturbations in opposite directions to produce responses with equal magnitude and opposite signs. Under this constraint, the mirrored branches maintain symmetric responses, thereby reducing the accumulated influence of stochastic perturbations. Meanwhile, the persistent degradation bias is not canceled and is further transformed into a discernible state discrepancy.

Dynamic Update Process The state evolution is implemented by a semi-implicit Euler scheme to obtain a stable discrete approximation of the second-order dynamics (Hairer et al., 2006). Random initialization is adopted to reduce the dependence on a specific initial state. At each time step t , the system computes the instantaneous acceleration $\ddot{\mathbf{x}}^{(t)}$ based on the current state. The update equations for the coupled branches are:

$$\ddot{\mathbf{x}}_L^{(t)} = -\gamma \dot{\mathbf{x}}_L^{(t)} - \mathcal{N}_r(\mathbf{x}_L^{(t)}) - \mathbf{\Gamma}_L^{(t)} + A \sin(\omega t), \quad (9)$$

$$\ddot{\mathbf{x}}_M^{(t)} = -\gamma \dot{\mathbf{x}}_M^{(t)} - \mathcal{N}_l(\mathbf{x}_M^{(t)}) - \mathbf{\Gamma}_M^{(t)} + C_M \mathcal{R}(t), \quad (10)$$

$$\ddot{\mathbf{x}}_R^{(t)} = -\gamma \dot{\mathbf{x}}_R^{(t)} - \mathcal{N}_r(\mathbf{x}_R^{(t)}) - \mathbf{\Gamma}_R^{(t)} + A \sin(\omega t + \pi). \quad (11)$$

Subsequently, the system first updates velocity and then advances position with the updated velocity:

$$\dot{\mathbf{x}}_{\{L,M,R\}}^{(t+1)} = \dot{\mathbf{x}}_{\{L,M,R\}}^{(t)} + \ddot{\mathbf{x}}_{\{L,M,R\}}^{(t)} \cdot \Delta t \quad (12)$$

$$\mathbf{x}_{\{L,M,R\}}^{(t+1)} = \mathbf{x}_{\{L,M,R\}}^{(t)} + \dot{\mathbf{x}}_{\{L,M,R\}}^{(t+1)} \cdot \Delta t \quad (13)$$

Through this iterative evolution under the coupled dynamics, directionally unstable random noise is unlikely to accumulate within the anti-symmetric structure, whereas persistent

shifts induced by degradation are gradually reflected in the deviation between the left and right branches, yielding the bilateral discrepancy energy $\mathbf{E}$:

$$\mathbf{E} = \sum_{t=1}^T \|\mathbf{x}_L^{(t)} + \mathbf{x}_R^{(t)}\|_2 \quad (14)$$

The energy metric $\mathbf{E}$ is then mapped to the battery lifetime through a linear projection. Appendix C summarizes the full procedure in pseudocode.

5. Experiments

We evaluate DITING on various battery datasets under the standardized BatteryLife (Tan et al., 2025) benchmark. We first report the experimental setup and overall comparison results (Sections 5.1 and 5.2). We then present ablation and component analyses to validate model behavior under different designs (Section 5.3). We further report cross-dataset zero-shot prediction results (Section 5.4) and examine performance trends with fewer input cycles and efficiency costs (Sections 5.5 and 5.6). Code is available at: <https://github.com/PHM-Batt/DITING>.

5.1. Experimental Setup

Datasets The experiments cover eight publicly available battery datasets, including CALCE, SNL, HNEI, HUST, XJTU, CALB, MATR, and MICH. These datasets collectively span diverse cell formats, nominal capacities, cycling protocols, operating temperatures, and data-acquisition settings across controlled laboratory tests and industrial aging scenarios. Dataset statistics and detailed descriptions, together with the original source references for each dataset, are provided in Appendix D.1.

Data Preprocessing Following recent battery lifetime benchmarks (Severson et al., 2019; Tan et al., 2025), each battery is represented by its first 100 cycles, with zero padding applied to shorter sequences. To reduce inconsistencies caused by varying sampling frequencies and cycle lengths, we adopt a unified preprocessing protocol: each cycle is standardized through segment-wise resampling and a consistent normalization scheme, while incomplete trajectories are processed under the same rules. Ground-truth lifetime labels are defined as the cycle index at which the state-of-health first falls below a prescribed threshold, using an 80% criterion for most datasets and a 90% criterion for CALB due to its testing standard, consistent with BatteryLife (Tan et al., 2025). Additional filtering and boundary extrapolation are applied to reduce label noise induced by truncated tests. The unified preprocessing protocol is detailed in Appendix D.2.

Table 1. Overall performance in terms of MAPE. Best values are highlighted in red and second-best values are underlined in blue.

DATASETS	DITING (OURS)	CPMLP (2025)	CPTRANSFORMER (2025)	CPGRU (2025)	LiPM (2025)	BATLiNET (2025)	DIFFBATT (2024)	iTRANSFORMER (2023)
CALCE	0.187 ± 0.005	<u>0.221</u> ± 0.004	0.226 ± 0.007	0.247 ± 0.004	0.257 ± 0.008	0.233 ± 0.007	0.225 ± 0.009	0.256 ± 0.010
SNL	0.152 ± 0.002	0.208 ± 0.003	<u>0.174</u> ± 0.007	0.309 ± 0.009	0.204 ± 0.006	0.221 ± 0.006	0.178 ± 0.007	0.229 ± 0.012
HNEI	0.035 ± 0.001	<u>0.036</u> ± 0.001	<u>0.036</u> ± 0.001	0.037 ± 0.001	0.035 ± 0.001	0.038 ± 0.002	0.037 ± 0.002	0.038 ± 0.002
HUST	<u>0.117</u> ± 0.003	0.139 ± 0.006	0.138 ± 0.004	0.098 ± 0.006	0.146 ± 0.006	0.149 ± 0.013	0.167 ± 0.008	0.153 ± 0.008
XJTU	0.049 ± 0.002	<u>0.055</u> ± 0.004	0.063 ± 0.004	0.076 ± 0.006	0.062 ± 0.004	0.073 ± 0.009	0.077 ± 0.006	0.080 ± 0.011
CALB	<u>0.169</u> ± 0.004	0.148 ± 0.003	0.175 ± 0.006	0.195 ± 0.014	0.177 ± 0.012	0.278 ± 0.018	0.256 ± 0.011	0.176 ± 0.023
MATR	0.125 ± 0.005	0.112 ± 0.006	<u>0.103</u> ± 0.003	0.097 ± 0.002	0.132 ± 0.007	0.158 ± 0.012	0.144 ± 0.014	0.149 ± 0.008
MICH	0.102 ± 0.007	<u>0.119</u> ± 0.004	0.124 ± 0.005	0.120 ± 0.004	0.155 ± 0.006	0.204 ± 0.012	0.179 ± 0.010	0.163 ± 0.006

Table 2. Overall performance in terms of 15%-ACC. Best values are highlighted in red and second-best values are underlined in blue.

DATASETS	DITING (OURS)	CPMLP (2025)	CPTRANSFORMER (2025)	CPGRU (2025)	LiPM (2025)	BATLiNET (2025)	DIFFBATT (2024)	iTRANSFORMER (2023)
CALCE	0.500 ± 0.000	0.500 ± 0.000	0.500 ± 0.000	0.500 ± 0.000	0.500 ± 0.000	0.500 ± 0.000	0.500 ± 0.000	0.500 ± 0.000
SNL	0.671 ± 0.014	0.434 ± 0.016	0.458 ± 0.026	0.507 ± 0.022	0.510 ± 0.019	0.462 ± 0.023	<u>0.598</u> ± 0.025	0.496 ± 0.025
HNEI	1.000 ± 0.000	1.000 ± 0.000	1.000 ± 0.000	1.000 ± 0.000	1.000 ± 0.000	1.000 ± 0.000	1.000 ± 0.000	1.000 ± 0.000
HUST	0.760 ± 0.018	0.576 ± 0.022	0.585 ± 0.018	<u>0.687</u> ± 0.019	0.645 ± 0.017	0.602 ± 0.035	0.655 ± 0.030	0.548 ± 0.019
XJTU	<u>0.911</u> ± 0.012	0.929 ± 0.017	0.810 ± 0.020	0.779 ± 0.012	0.875 ± 0.010	0.705 ± 0.022	0.812 ± 0.026	0.750 ± 0.018
CALB	0.625 ± 0.047	<u>0.656</u> ± 0.051	0.616 ± 0.089	0.600 ± 0.135	0.684 ± 0.189	0.503 ± 0.203	0.511 ± 0.175	0.612 ± 0.058
MATR	<u>0.733</u> ± 0.019	0.715 ± 0.014	0.726 ± 0.017	0.778 ± 0.010	0.669 ± 0.021	0.602 ± 0.028	0.630 ± 0.022	0.626 ± 0.012
MICH	0.750 ± 0.000	0.750 ± 0.000	0.750 ± 0.000	0.750 ± 0.000	0.750 ± 0.000	0.750 ± 0.000	0.750 ± 0.000	0.750 ± 0.000

Evaluation Metrics Two metrics are selected to evaluate lifetime prediction accuracy under the same evaluation protocol: Mean Absolute Percentage Error (MAPE) and threshold-based accuracy within a 15% tolerance (15%-ACC). Let y_i denote the ground-truth label, $\hat{y}_i$ denote the predicted lifetime for the i -th test sample, and N denote the number of test samples. MAPE evaluates the relative prediction error normalized by the ground-truth lifetime:

$$\text{MAPE} = \frac{1}{N} \sum_{i=1}^N \left| \frac{y_i - \hat{y}_i}{y_i} \right| \times 100\%. \quad (15)$$

In addition, 15%-ACC measures the proportion of predictions with relative error within the 15% tolerance of the ground-truth lifetime:

$$15\text{-}\text{ACC} = \frac{1}{N} \sum_{i=1}^N \mathbb{I} \left[|y_i - \hat{y}_i| \leq 0.15 \cdot y_i \right], \quad (16)$$

where $\mathbb{I}[\cdot]$ denotes the indicator operator.

5.2. Experimental Results

We compare DITING with representative lifetime prognostics methods, including CPMLP, CPGRU, CPTransformer, DiffBatt, BatLiNet, LiPM, and iTransformer. All methods are evaluated under the same runtime environment with ten independent runs. We report the mean and standard deviation to reflect performance stability under different random initializations. Detailed descriptions of the baseline methods are provided in Appendix D.3, and the experimental environment and training details are provided in Appendix D.4.

Tables 1 and 2 report the overall results on eight datasets. DITING achieves the best or tied-best result on 10 of the 16 reported dataset–metric pairs and attains the second-best result on four additional pairs. In particular, DITING obtains strong performance on CALCE, SNL, HNEI, XJTU, and MICH, showing stable accuracy across different testing protocols. The results also indicate that DITING is not uniformly dominant on every dataset and metric. For example, larger protocol variation in HUST and MATR appears to favor recurrent inductive bias, which benefits CPGRU. Nevertheless, DITING remains competitive over the complete benchmark, supporting the effectiveness of manifesting weak degradation cues from noisy early cycles.

5.3. Ablation Studies

We conduct ablation studies on five datasets to assess the contribution of each component and further disentangle the mechanisms in the bilateral dynamics. Six variants are constructed, covering component-level removals and targeted mechanism-level modifications to the bilateral response formulation under controlled settings:

- **w/o-1:** Remove hierarchical embedding and use the cycle representation directly for health prototype construction and tri-coupled degradation manifestation.
- **w/o-2:** Remove health prototype construction and adopt the first cycle as the reference for computing residual degradation evidence.
- **w/o-3:** Remove tri-coupled degradation manifestation and use the residual representation for lifetime prediction without discrepancy-aware manifestation.

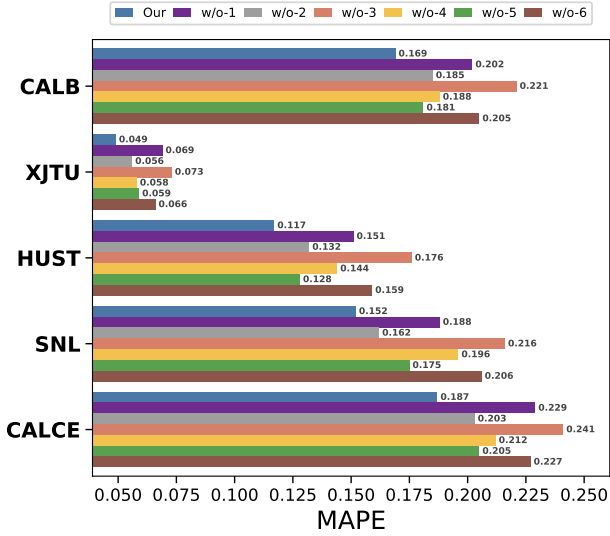


Figure 4. Ablation results of DITING on five datasets by MAPE, including component-level and bilateral-mechanism variants.

- **w/o-4:** Replace the anti-phase drive with an in-phase drive while keeping the remaining components unchanged.
- **w/o-5:** Remove the symmetry constraint by replacing the symmetrized response $\mathcal{N}(x) = \phi(x) - \phi(-x)$ with a standard nonlinear mapping $\phi(x)$, while retaining the anti-phase drive.
- **w/o-6:** Apply both modifications in w/o-4 and w/o-5 to test the joint contribution of anti-phase driving and symmetrized response.

Figure 4 presents the ablation results on five representative datasets. The ablation variants generally lead to performance degradation, indicating the stable contribution of the full design to early lifetime prediction. At the component level, removing tri-coupled degradation manifestation causes the most pronounced degradation, showing that discrepancy-aware manifestation is critical for extracting weak degradation cues under stochastic fluctuations. Replacing health prototype construction with a first-cycle reference also degrades performance, suggesting that a robust health anchor helps prevent single-cycle noise from entering the degradation representation. Removing hierarchical embedding further reduces performance, indicating that multi-level representations stabilize early-cycle features before degradation manifestation and lifetime prediction. At the mechanism level, the results of w/o-4 and w/o-5 show that both the anti-phase drive and the symmetry constraint contribute to the bilateral dynamics. Removing the anti-

Table 3. Zero-shot lifetime prediction under leave-one-dataset-out training. Best values are highlighted in red and second-best values are underlined in blue.

HELD-OUT TARGET	DITING (Ours)	CPMLP (2025)	LIPM (2025)	DIFFBATT (2024)	BATLiNET (2025)
CALCE	0.642	<u>0.695</u>	0.783	0.874	0.927
SNL	0.537	0.706	0.743	<u>0.693</u>	0.851
HNEI	<u>0.137</u>	0.176	0.123	0.244	0.254
HUST	0.364	<u>0.469</u>	0.533	0.619	0.658
XJTU	0.175	<u>0.238</u>	0.276	0.382	0.335
CALB	<u>0.538</u>	0.420	0.677	0.894	0.662
MATR	<u>0.312</u>	0.286	0.381	0.427	0.469
MICH	0.294	<u>0.318</u>	0.337	0.386	0.431

phase drive usually causes a larger degradation, suggesting that it more directly establishes the bilateral comparison frame. Removing the symmetry constraint also weakens performance, indicating that this constraint helps suppress directionally inconsistent perturbations from forming spurious discrepancies. When both mechanisms are removed, the degradation becomes the most evident, further showing that the bilateral mechanism depends on the interaction between the anti-phase drive and the symmetry constraint.

5.4. Zero-shot Prediction

We further evaluate zero-shot generalization under cross-dataset shifts. For each target dataset, the model is trained on the union of training sets from the other seven datasets and tested on the held-out target without using any target-domain labels. This protocol yields eight zero-shot settings. Table 3 reports MAPE for representative lifetime prediction methods. Compared with the in-domain evaluation, all methods incur larger errors in the cross-dataset setting, indicating that differences in aging conditions, testing protocols, and measurement environments substantially increase the difficulty of early lifetime prediction. DITING remains strongly competitive in the zero-shot setting, suggesting that the bilateral discrepancy representation can extract relatively stable degradation evidence from early-cycle observations. In contrast, some baselines exhibit larger performance drops when the health reference and degradation patterns shift across domains, indicating higher sensitivity to dataset-specific patterns. These results suggest that improving cross-dataset robustness requires not only stronger in-domain fitting, but also more stable extraction of transferable degradation evidence from early cycles.

5.5. Prediction with Fewer Cycles

To examine early prediction under restricted observation windows, we vary the input horizon to 20, 40, 60, 80, and 100 cycles and report MAPE. Figure 5 shows the performance trend on CALCE for DITING and CPMLP, illustrating how prediction error changes as early-cycle evidence increases. When fewer cycles are observed, errors increase for both methods, indicating that shorter input windows

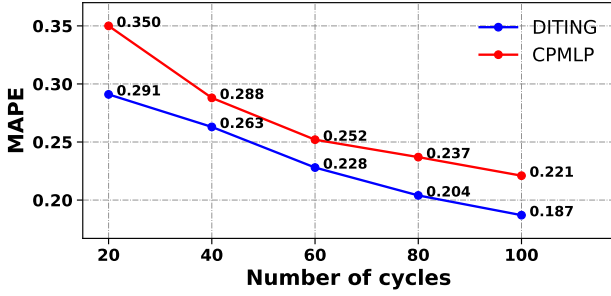


Figure 5. Prediction with fewer cycles. MAPE on CALCE for DITING and CPMLP as the number of observed cycles varies from 20 to 100.

provide less degradation information for lifetime prediction. Nevertheless, DITING consistently achieves lower MAPE than CPMLP across all horizons, suggesting that its bilateral discrepancy representation preserves more informative degradation cues when early evidence is limited. Appendix D.5 reports the complete results on other datasets and further provides a noise injection experiment to examine the stability of the extracted degradation evidence under measurement perturbations.

5.6. Efficiency Analysis

Figure 6 compares training time and memory consumption across representative baselines under the same setting, where bubble size indicates the parameter count. The results place DITING among methods that achieve high throughput under moderate memory usage, while avoiding the substantially higher resource footprint observed for diffusion models and foundation models. Compared with the predictive baselines, DITING retains a lightweight computational profile yet attains a more favorable balance between accuracy and efficiency than solutions that predominantly improve performance by scaling parameters. Overall, the proposed design improves lifetime prediction with limited additional overhead, supporting its use in early-cycle settings with constrained compute and memory budgets.

6. Conclusion

In this work, we propose DITING, a bio-inspired weak degradation listener that establishes a discrepancy-aware paradigm for battery lifetime early prediction. By constructing a robust health prototype via Sinkhorn optimal transport and introducing a tri-coupled degradation manifestation mechanism, DITING enables subtle aging cues from early cycles to be manifested through bilateral discrepancy while reducing the interference of stochastic fluctuations in degradation representation. Extensive experiments demonstrate

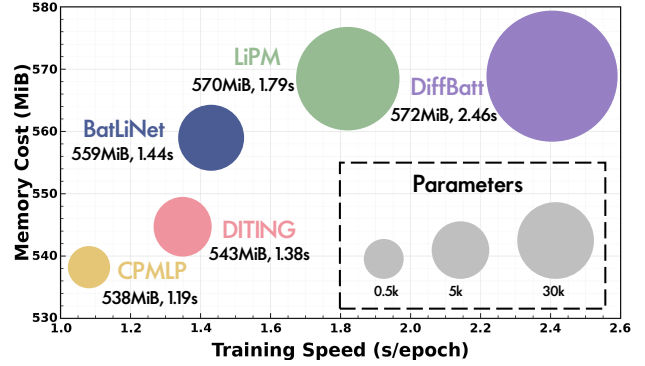


Figure 6. Training time versus memory consumption across representative baselines.

that our approach achieves strong prediction accuracy and robust generalization across diverse aging conditions. Future work will further explore the generalizability of this discrepancy-aware perception framework across broader time-series analysis tasks, investigate more lightweight implementations, and theoretically study the interaction between nonlinear coupling dynamics and complex noise distributions to further push the robustness boundaries of weak signal detection.

Acknowledgements

This work was supported by the Joint Fund for Regional Innovation and Development of NSFC (No. U22A20167) and the Special Project for Guiding the Transformation of Scientific and Technological Achievements in Shanxi Province (No. 202404021301033).

Impact Statement

This work addresses a machine learning challenge, extracting weak degradation cues from noisy battery observations under real-world conditions. Its bio-inspired design moves beyond simple denoising by turning persistent degradation deviations into usable representations while reducing stochastic interference. This can improve early lifetime estimation, safety assessment, and lifecycle management for batteries in electric mobility and renewable-grid systems. More broadly, the same principle may benefit other noisy weak-signal settings, including physiological monitoring, industrial sensing, and remote observation, where subtle patterns are often hidden by noise. This paper presents work whose goal is to advance the field of machine learning. There are many potential societal consequences of our work, none of which we feel must be specifically highlighted here.

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A. Motivation: Discrepancy-Aware Perception from the Binaural Effect

Detecting weak cues under strong interference is a shared challenge in biological perception and early-stage battery lifetime prediction. In complex acoustic environments, auditory systems must perceive informative cues under competing sources, reverberation, and stochastic perturbations. The binaural effect shows that many organisms do not rely only on a single observation stream. Instead, they improve perceptual reliability by comparing two correlated sensory pathways. The key role of this mechanism is not to simply amplify signal magnitude, but to exploit structurally meaningful differences between bilateral inputs, making weak cues more stable under interference.

From a signal processing perspective, binaural perception mainly relies on the interaural time difference (ITD) and the interaural level difference (ILD). ITD describes the relative temporal delay and phase difference between the left and right pathways, while ILD describes the energy difference induced by head-related shadowing and propagation attenuation. These differences are not arbitrary noise. They are structured cues associated with the propagation geometry of the sound source. Therefore, binaural processing can be understood as a comparison and integration mechanism. Directionally unstable or approximately symmetric perturbations are unlikely to form persistent bilateral discrepancies, whereas informative cues with a consistent bias become more pronounced after bilateral comparison.

This mechanism provides a functional inspiration for early battery lifetime prediction. We emphasize that DITING does not assume that battery measurements naturally contain two physical sensory streams, nor does it treat the binaural effect as a physical counterpart of battery degradation. Instead, we borrow its discrepancy-aware principle and construct two mirrored latent dynamical responses from the same degradation residual. Early degradation signatures are usually weak and entangled with measurement fluctuations, operating-condition variations, and sampling perturbations. Directly applying aggressive denoising or explicit decomposition may also attenuate weak degradation cues that are relevant to lifetime. Therefore, the objective of DITING is not to completely remove noise, but to reduce the chance that random perturbations are converted into stable degradation representations while facilitating the manifestation of directionally consistent degradation changes. Guided by the binaural effect, we abstract discrepancy-aware perception into three design principles in DITING:

- **Construct comparable bilateral responses:** DITING first extracts degradation residuals based on a health prototype and constructs two comparable latent responses from the same residual input, providing a unified health reference for bilateral discrepancy modeling.
- **Impose explicit symmetry constraints:** DITING introduces mirrored structures and opposite driving directions in the bilateral dynamics, making approximately zero-mean and directionally unstable random perturbations less likely to be rectified into stable spurious discrepancies by model non-idealities.
- **Use bilateral discrepancy rather than single-branch magnitude as the representation:** The magnitude of a single pathway can respond to both noise and degradation, whereas bilateral discrepancy focuses on whether there exists a persistent bias that breaks the mirrored equilibrium. Random perturbations usually lack a consistent direction, so their contribution is weakened under symmetric responses. Degradation changes are more likely to persistently deviate from the health reference and appear as stable separation between the left and right responses.

Concretely, DITING first constructs a health prototype and extracts degradation residuals. The residual is then injected into a central branch and coupled to two side branches with mirrored structures. The left and right branches receive opposite driving directions to form controlled bilateral responses. Since random noise is usually directionally unstable, its influence is less likely to accumulate into a stable bilateral discrepancy in expectation. In contrast, degradation changes are more directionally consistent, persistently break the mirrored equilibrium, and are manifested in the discrepancy energy. In this way, DITING translates the discrepancy-aware principle of biological perception into an early-cycle degradation representation mechanism, which underpins the design of the Tri-Coupled Degradation Manifestation module.

B. Analysis of Noise Suppression and Signal Manifestation

This section analyzes the noise suppression and signal manifestation mechanism in DITING. We first examine the symmetric response of the dynamical system, showing how mirrored side branches are maintained under opposite driving. We then analyze noise rectification, explaining why directionally unstable perturbations are less likely to accumulate into stable discrepancy energy. Finally, we discuss signal manifestation, where persistent degradation changes repeatedly break the mirrored balance and appear as stable bilateral discrepancies.

B.1. Symmetric Response of the Dynamical System

We consider the response of the two side branches in the tri coupled dynamical system. The symmetrized nonlinear response $\mathcal{N}(\mathbf{x})$ is defined by the central odd symmetry condition:

$$\mathcal{N}(-\mathbf{x}) = -\mathcal{N}(\mathbf{x}). \quad (17)$$

This condition means that mirrored states should induce opposite nonlinear responses. It does not imply that noise is removed point by point. Instead, it provides a structural constraint that helps prevent random perturbations from being rectified into a stable bias.

Let the input residual be

$$\mathcal{R}(t) = \mathbf{d}(t) + \mathbf{n}(t), \quad (18)$$

where $\mathbf{d}(t)$ denotes the directionally consistent degradation deviation from the health reference, and $\mathbf{n}(t)$ denotes a zero-mean stochastic perturbation. For one side branch, we write the response equation as

$$\begin{aligned} \mathcal{L}[\mathbf{x}](t) &= \mathbf{F}(t), \\ \mathcal{L}[\mathbf{x}](t) &\triangleq \ddot{\mathbf{x}}(t) + \gamma \dot{\mathbf{x}}(t) + \mathcal{N}(\mathbf{x}(t)). \end{aligned} \quad (19)$$

Here, γ is the damping coefficient, and $\mathbf{F}(t)$ denotes the driving force induced by residual coupling from the central branch.

When $\mathcal{N}$ is odd, the operator $\mathcal{L}$ satisfies the reflection relation:

$$\mathcal{L}[-\mathbf{x}](t) = -\mathcal{L}[\mathbf{x}](t). \quad (20)$$

Therefore, if $\mathbf{x}(t)$ is a response under $\mathbf{F}(t)$, then $-\mathbf{x}(t)$ is a response under $-\mathbf{F}(t)$. For the two side branches, this implies that opposite driving forces lead to mirrored responses under matched initial conditions. Specifically, when $\mathbf{F}_L(t) = \mathbf{F}(t)$, $\mathbf{F}_R(t) = -\mathbf{F}(t)$, $\mathbf{x}_R(0) = -\mathbf{x}_L(0)$, and $\dot{\mathbf{x}}_R(0) = -\dot{\mathbf{x}}_L(0)$, the symmetric response component satisfies

$$\mathbf{x}_R(t) = -\mathbf{x}_L(t). \quad (21)$$

This mirrored response forms the basis of the noise suppression effect. Since zero-mean perturbations have no stable direction, they are less likely to accumulate into a persistent signed discrepancy under the symmetric constraint. If the odd symmetry is broken, even order terms in the nonlinear response may produce the same sign of response for positive and negative perturbations. Such terms can convert zero-mean noise into a nonzero drift, leading to spurious stable discrepancy energy. By contrast, degradation changes are not directionless fluctuations. They represent persistent deviations from the health reference, so they can repeatedly break the mirrored balance and be reflected in the discrepancy energy.

B.2. Noise Rectification Analysis

Theorem B.1. *Under the symmetric response constraint, if the response satisfies $\mathcal{N}(-\mathbf{x}) = -\mathcal{N}(\mathbf{x})$, then zero-mean stochastic perturbations introduce no expected signed drift in the discrepancy response under the first-order perturbation view. If the response contains even order components, zero-mean perturbations may be rectified into a nonzero drift.*

Proof. Consider the equations of motion for the left and right reference branches. The left branch is driven by $F_{\text{drive}}(t)$, and the right branch is driven by $-F_{\text{drive}}(t)$. The middle branch provides perturbations induced by the input residual. Since the response function $\mathcal{N}$ is odd, the dynamical operator $\mathcal{L}[\mathbf{x}] = \ddot{\mathbf{x}} + \gamma \dot{\mathbf{x}} + \mathcal{N}(\mathbf{x})$ satisfies reflection equivariance. This implies that if $\mathbf{x}(t)$ is a solution under the driving force $\mathbf{F}(t)$, then $-\mathbf{x}(t)$ is a solution under the opposite driving force $-\mathbf{F}(t)$. More precisely, if $\mathcal{L}[\mathbf{x}] = \mathbf{F}$, then the oddness of $\mathcal{N}$ gives $\mathcal{L}[-\mathbf{x}] = -\mathcal{L}[\mathbf{x}] = -\mathbf{F}$. Therefore, the deterministic responses of the two side branches form a reflected solution pair. Under standard stability assumptions for dissipative nonlinear systems, the long time response is unique for a given input, so the right branch baseline matches the reflected left branch baseline.

For noise $\mathbf{n}(t)$ propagating from the middle branch, we regard it as a symmetric perturbation to both side branches. Let $\mathbf{x}_{\text{base}}(t)$ be the baseline response of the left branch to the deterministic driving force. The baseline response of the right branch is then $-\mathbf{x}_{\text{base}}(t)$. Let the perturbation induced by noise be $\delta\mathbf{x}(t)$. The states of the two branches can be written as

$$\mathbf{x}_L(t) = \mathbf{x}_{\text{base}}(t) + \delta\mathbf{x}(t), \quad \mathbf{x}_R(t) = -\mathbf{x}_{\text{base}}(t) + \delta\mathbf{x}(t). \quad (22)$$

To avoid the nonnegative bias introduced by an absolute value, we first analyze the signed energy discrepancy:

$$\Delta(t) = \|\mathbf{x}_L(t)\|^2 - \|\mathbf{x}_R(t)\|^2 = 4\langle \mathbf{x}_{\text{base}}(t), \delta \mathbf{x}(t) \rangle. \quad (23)$$

The signed drift vanishes when the response remains centrally odd. Under small noise, $\delta \mathbf{x}(t)$ can be analyzed by the first-order stochastic perturbation around $\mathbf{x}_{\text{base}}(t)$:

$$\delta \ddot{\mathbf{x}}(t) + \gamma \delta \dot{\mathbf{x}}(t) + \mathbf{J}(t) \delta \mathbf{x}(t) = \mathbf{G} \mathbf{n}(t), \quad (24)$$

where $\mathbf{J}(t) \triangleq \frac{\partial \mathcal{N}}{\partial \mathbf{x}}|_{\mathbf{x}_{\text{base}}(t)}$ is the Jacobian along the baseline trajectory, and $\mathbf{G}$ denotes the symmetric coupling pathway from the middle branch to the two side branches. When $\mathcal{N}$ is odd, its Jacobian is even. Thus, the linearized perturbation dynamics are consistent under the reflected baselines of the two side branches. With zero-mean input noise and a stable local response, the perturbation has no mean shift, namely $\mathbb{E}[\delta \mathbf{x}(t)] = \mathbf{0}$. Therefore,

$$\mathbb{E}[\Delta(t)] = 4 \langle \mathbf{x}_{\text{base}}(t), \mathbb{E}[\delta \mathbf{x}(t)] \rangle = 0. \quad (25)$$

This result means that the symmetric response suppresses the signed drift caused by zero-mean noise. It does not imply that the absolute discrepancy energy is zero. The absolute energy may still be nonzero because it measures fluctuation magnitude under finite perturbations in the coupled system.

If the response contains a small even order component, the above symmetry no longer holds. For example, let $\mathcal{N}(\mathbf{x}) = k\mathbf{x} + \epsilon \mathbf{x}^2$. For positive and negative perturbations, the even order term produces responses with the same sign and breaks the reflected relation between the two branches. As a result, the perturbation distribution can become biased even when the input noise is zero-mean, leading to a nonzero expected signed discrepancy. More generally, any non odd component of $\mathcal{N}$ breaks the reflection relation $\mathcal{L}[-\mathbf{x}] = -\mathcal{L}[\mathbf{x}]$, so that zero-mean perturbations may be converted into a rectified drift. In DITING, the symmetric nonlinear response is constructed as

$$\mathcal{N}(\mathbf{x}) = \phi(\mathbf{x}) - \phi(-\mathbf{x}). \quad (26)$$

Under a local Taylor expansion of ϕ , this subtraction cancels even order terms and keeps only odd order terms:

$$\phi(\mathbf{x}) - \phi(-\mathbf{x}) = 2 \sum_{m=0}^{\infty} \mathbf{A}_{2m+1} \mathbf{x}^{2m+1}. \quad (27)$$

Therefore, the constructed response remains centrally odd and reduces the nonlinear pathway through which zero-mean perturbations are rectified into signed drift. This provides a sufficient condition for suppressing noise rectification under the stated perturbation assumptions. In DITING, the final discrepancy energy is used to measure the magnitude of manifested asymmetry. Hence it can remain nonzero for finite fluctuations, while the symmetric response reduces noise induced signed bias and allows persistent degradation deviations to break the mirrored balance more consistently.

B.3. Signal Manifestation Analysis

Theorem B.2. *Under the mirrored response structure, a degradation deviation that remains directionally consistent relative to the health reference can induce a nonzero expected signed discrepancy between the two side branches. This discrepancy can then be reflected by the nonnegative discrepancy energy used for prediction.*

Proof. Unlike stochastic noise, the degradation signal $\mathbf{d}(t)$ is not a directionless fluctuation. It represents a persistent deviation from the health reference and can act as a coherent input to the middle branch. To describe its effect, we define the accumulated work-like contribution

$$W_{\text{deg}}(T) = \int_0^T \mathbf{d}(t) \cdot \dot{\mathbf{x}}_{\text{M}}(t) dt. \quad (28)$$

For zero-mean noise $\mathbf{n}(t)$ that has no persistent alignment with the middle branch velocity, the expected time-averaged injected power vanishes:

$$\mathbb{E} \left[\frac{1}{T} \int_0^T \mathbf{n}(t) \cdot \dot{\mathbf{x}}_{\text{M}}(t) dt \right] = 0. \quad (29)$$

Table 4. Statistical information of the battery datasets.

DATASETS	CALCE	SNL	HNEI	HUST	XJTU	CALB	MATR	MICH
BATTERY NUMBER	13	52	14	77	23	27	169	40
FORMAT	1	1	1	1	1	1	1	1
CHEMICAL SYSTEM	1	3	1	1	1	1	1	1
OPERATION TEMPERATURE	1	3	1	1	1	4	1	2
PROTOCOL	2	23	2	77	2	2	81	2
DATA SOURCE	LAB TEST	LAB TEST	LAB TEST	LAB TEST	LAB TEST	INDUSTRIAL TEST	LAB TEST	LAB TEST
BATTERY TYPE	LI-ION	LI-ION	LI-ION	LI-ION	LI-ION	LI-ION	LI-ION	LI-ION

By contrast, when $\mathbf{d}(t)$ keeps a consistent direction along the degradation process, it can maintain a nonzero long time correlation with $\dot{\mathbf{x}}_M(t)$, namely

$$\lim_{T \rightarrow \infty} \frac{1}{T} \int_0^T \mathbf{d}(t) \cdot \dot{\mathbf{x}}_M(t) dt = \rho_{\text{deg}}, \quad \rho_{\text{deg}} \neq 0. \quad (30)$$

This condition does not require the degradation force to dominate the dynamics. It only states that degradation can introduce a persistent bias, while zero-mean random perturbations do not have such stable alignment in expectation. The biased middle branch response is then transferred to the two side branches through coupling. The effective driving forces can be written as

$$\mathbf{F}_L^{\text{eff}}(t) = \mathbf{F}_{\text{drive}}(t) + \kappa \mathbf{C} \mathbf{x}_M(t), \quad \mathbf{F}_R^{\text{eff}}(t) = -\mathbf{F}_{\text{drive}}(t) + \kappa \mathbf{C} \mathbf{x}_M(t), \quad (31)$$

where κ denotes the coupling strength, and $\mathbf{C}$ denotes the coupling projection. Without degradation, the two side branches follow mirrored responses under opposite driving. With a persistent degradation-induced shift in $\mathbf{x}_M(t)$, the same coupling term is superposed onto two opposite baseline drives. Therefore, the left and right branches move to different effective operating points. This breaks the mirrored balance and creates a systematic energy discrepancy. Let $E_L(t)$ and $E_R(t)$ denote the branch energies. We define the long time signed energy discrepancy as

$$\Delta E(T) = \frac{1}{T} \int_0^T (E_L(t) - E_R(t)) dt. \quad (32)$$

In the absence of a coherent degradation bias, the mirrored response gives $\mathbb{E}[\Delta E(T)] = 0$ under the symmetry condition. When the degradation deviation induces a persistent middle branch bias, the coupling term changes the work exchange of the two side branches differently, so the signed discrepancy can satisfy

$$\mathbb{E}[\Delta E(T)] \neq 0. \quad (33)$$

The final discrepancy energy used by DITING is a nonnegative measure of this manifested asymmetry. Hence, it shows that persistent degradation deviations can repeatedly break the mirrored equilibrium and become visible through the accumulated discrepancy energy.

C. DITING Algorithm

This section provides the pseudocode of DITING. Algorithm 1 summarizes the complete procedure, including health prototype construction, tri-coupled degradation manifestation, and lifetime prediction from discrepancy energy.

D. Experimental Details

D.1. Benchmark Datasets and Sources

This work adopts eight public battery datasets consolidated by the BatteryLife benchmark (Tan et al., 2025). These datasets cover diverse battery chemistries, cycling protocols, operating conditions, and lifetime ranges, providing a broad testbed for early lifetime prediction under heterogeneous degradation behaviors. To ensure comparability across datasets, we follow the unified dataset reporting and standardized preprocessing protocol of BatteryLife. For each dataset, we also cite its original public paper or data release to clarify the data source. Table 4 summarizes the dataset statistics.

CALCE. CALCE provides prismatic Li-ion cells tested under two cycling protocols at a fixed temperature. It is a compact but widely used benchmark for early battery lifetime prediction, with its prognostic use documented in (Xing et al., 2013).

Algorithm 1 DITING: Tri-Coupled Manifestation with Discrete-Time Updates

```

1: Input: early-cycle measurements  $U_{1:T}$ , lifetime label  $Y$  for training
2: Output: predicted lifetime  $\hat{Y}$ 
3: Learnable modules: embedding  $\Phi$ , prototype builder  $\mathcal{G}$ , nonlinear fields  $\mathcal{N}_r, \mathcal{N}_l$ , linear head  $(w, b)$ 
4: Hyperparameters: coupling  $\kappa$ , damping  $\gamma$ , drive amplitude  $A$ , frequency  $\omega$ , injection gain  $C_M$ , step size  $\Delta t$ 

5: // (1) Embedding and prototype
6: for  $t = 1$  to  $T$  do
7:    $x_t \leftarrow \Phi(U_t)$ 
8: end for
9:  $H \leftarrow \mathcal{G}(x_{1:K})$  // Sinkhorn OT selective matching and weighted fusion
10: for  $t = 1$  to  $T$  do
11:    $r_t \leftarrow x_t - H$ 
12: end for
13: // (2) Tri-coupled discrete-time manifestation
14: Initialize states  $(x_L, x_M, x_R)$  and velocities  $(v_L, v_M, v_R)$ 
15:  $E \leftarrow 0$ 
16: for  $t = 1$  to  $T$  do
17:    $f_t \leftarrow A \sin(\omega t)$  // anti-phase drive
18:    $a_L \leftarrow -\gamma v_L - \mathcal{N}_r(x_L) - \kappa(x_L - x_M) + f_t$ 
19:    $a_M \leftarrow -\gamma v_M - \mathcal{N}_l(x_M) - \kappa(2x_M - x_L - x_R) + C_M r_t$ 
20:    $a_R \leftarrow -\gamma v_R - \mathcal{N}_r(x_R) - \kappa(x_R - x_M) - f_t$ 
21:    $v_L \leftarrow v_L + a_L \Delta t, \quad v_M \leftarrow v_M + a_M \Delta t, \quad v_R \leftarrow v_R + a_R \Delta t$ 
22:    $x_L \leftarrow x_L + v_L \Delta t, \quad x_M \leftarrow x_M + v_M \Delta t, \quad x_R \leftarrow x_R + v_R \Delta t$ 
23:    $E \leftarrow E + \|x_L + x_R\|_2$  // bilateral discrepancy energy
24: end for
25: // (3) Prediction and training
26:  $\hat{Y} \leftarrow wE + b$ 
27: Minimize the mean squared error between  $\hat{Y}$  and  $Y$  to update all parameters end-to-end.
    
```

HNEI. HNEI includes commercial 18650 Li-ion cells tested under controlled laboratory conditions. It is useful for examining intrinsic cell-to-cell variability, and its original description is provided in (Devie et al., 2018).

SNL. SNL covers 18650 Li-ion cells with multiple cathode chemistries, operating temperatures, and cycling protocols. It provides a representative setting for evaluating robustness under chemistry and condition shifts (Preger et al., 2020).

HUST. HUST is a larger 18650 benchmark with diverse multi-stage discharge protocols under a fixed temperature. Its large protocol variation makes it suitable for evaluating prediction under protocol-induced distribution shift (Ma et al., 2022).

XJTU. XJTU provides recent cycling records of 18650 Li-ion cells under two operating conditions at a fixed temperature. It serves as a useful benchmark for early lifetime prediction with modern battery testing records (Wang et al., 2024a).

CALB. CALB includes industry-tested large-capacity prismatic Li-ion cells provided by CALB Group Co., Ltd. The dataset involves multiple operating temperatures and industrial protocols, with its standardized organization and test criteria documented by BatteryLife (Tan et al., 2025).

MATR. MATR provides commercial 18650 Li-ion cells tested under a large set of fast-charging protocols and identical discharge conditions. It is widely used for early lifetime prediction and is particularly useful for evaluating protocol-dependent degradation patterns (Severson et al., 2019; Attia et al., 2020).

MICH. MICH is designed to study how formation protocols affect battery lifetime immediately after manufacturing. It uses pouch-format Li-ion cells and provides an additional benchmark for evaluating early lifetime prediction under formation-protocol variations (Weng et al., 2021).

D.2. Unified Preprocessing Protocol

Raw cycling records differ in sampling frequency, cycle duration, and the number of time steps per cycle. We follow the unified preprocessing protocol of BatteryLife (Tan et al., 2025) and summarize the main steps below.

Data cleaning. Before constructing model inputs, abnormal cycles are filtered according to the BatteryLife preprocessing protocol. Rest performance tests and formation cycles are removed when they are not part of the regular cycling protocol, and cycles with sudden abnormal rises or drops in discharge capacity are excluded. Batteries whose life labels are no larger than 100 cycles are also removed, since our early prediction setting uses the first 100 cycles as input.

Cycle-wise resampling. Charging and discharging segments are each resampled to a fixed length of 150 using linear interpolation. Each cycle is therefore represented by 300 resampled points, with voltage, current, and capacity records aligned under a unified temporal resolution.

Input window and padding. The first 100 cycles are used as the early-life prefix, consistent with the BatteryLife benchmark setting. Batteries with fewer than 100 usable cycles after cleaning are padded with zeros to form a fixed-length input tensor.

Normalization. Variables are normalized to reduce scale differences while preserving intra-cycle structure. Following the benchmark convention, capacity and current are normalized by nominal capacity, voltage is normalized by its cycle-wise maximum, and the resampled variables are concatenated as the cycle representation.

Label construction and noise handling. Life labels are defined by the cycle index at which the state of health first falls below a threshold λ , following common practice in battery lifetime prediction (Severson et al., 2019; Tan et al., 2025). We use $\lambda = 80\%$ for CALCE, SNL, HNEI, HUST, XJTU, MATR, and MICH. For CALB, we use $\lambda = 90\%$ due to its test termination criterion, as documented by BatteryLife (Tan et al., 2025).

A subset of batteries may stop cycling before reaching the threshold, which can introduce label noise if the last recorded cycle is directly used as the end of life. To mitigate this issue, batteries whose final state of health remains above $\lambda + 2.5\%$ are excluded, while batteries whose final state of health lies between λ and $\lambda + 2.5\%$ are handled by linear extrapolation to estimate the threshold-crossing cycle.

D.3. Baseline Methods

This section summarizes the baseline methods used in Tables 1 and 2. All models are trained and evaluated with the same data splits, early-cycle inputs, preprocessing protocol, and evaluation metrics to ensure fair comparison.

CPMLP. CPMLP (Tan et al., 2025) first represents each cycle with patch-level features and then aggregates the early-cycle representations using a multilayer perceptron. It focuses on direct regression from structured cycle patches without explicit recurrent or attention-based temporal modeling.

CPTransformer. CPTransformer (Tan et al., 2025) builds on cycle patch representations and introduces a Transformer encoder to model dependencies among early cycles. The attention mechanism allows the model to capture long-range interactions across cycles while retaining intra-cycle patch information.

CPGRU. CPGRU (Tan et al., 2025) uses gated recurrent units to process cycle patch representations in chronological order. The recurrent structure models the evolution of early-cycle patterns and captures sequential degradation dependencies.

LiPM. LiPM (Li et al., 2025) is a pre-training based battery time-series model. It learns general representations from large-scale battery cycling data and adapts the learned features to downstream lifetime prediction.

BatLiNet. BatLiNet (Zhang et al., 2025) models lifetime prediction from an inter-cell learning perspective. It exploits relations among cells aged under different conditions, allowing information from related batteries to support the prediction of a target cell.

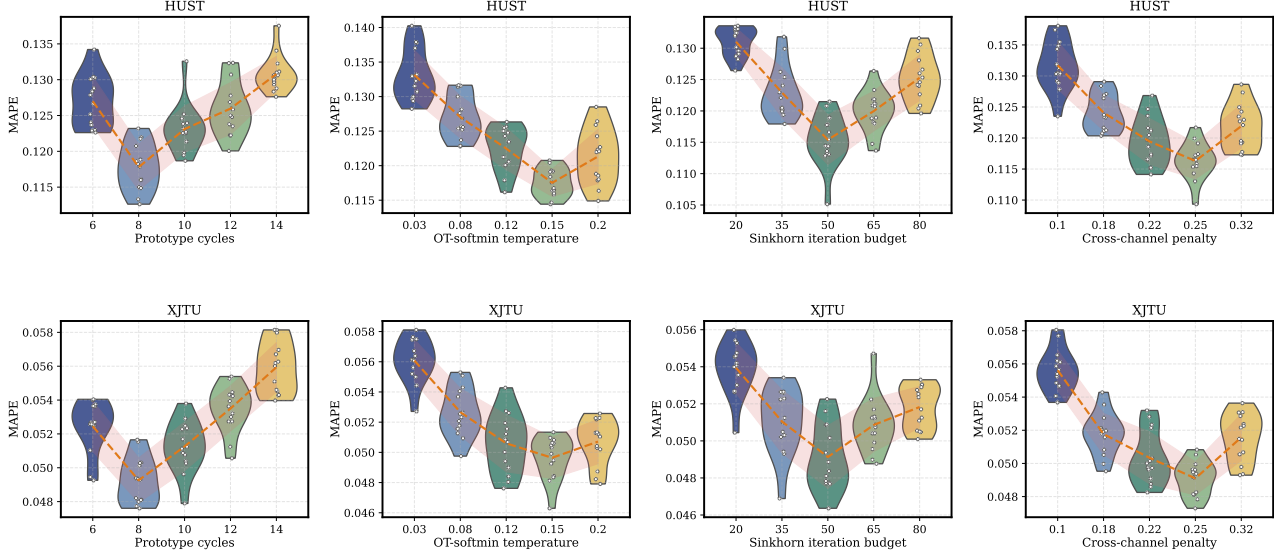


Figure 7. Sensitivity analysis of key DITING hyperparameters.

DiffBatt. DiffBatt (Eivazi et al., 2024) formulates battery trajectory modeling with a diffusion framework. It learns generative degradation representations from cycling data and uses them to enhance early-cycle lifetime prediction.

iTransformer. iTransformer (Liu et al., 2024) is a general time-series Transformer model that treats variables as tokens and applies attention across variable-wise representations. This design improves the modeling of multivariate dependencies and long-range temporal patterns.

D.4. Experimental Details

All experiments were conducted on a Linux server running Ubuntu 20.04, equipped with an NVIDIA GeForce RTX 4090 GPU with 24 GB memory and CUDA 12.8, an AMD Ryzen CPU at 3.5 GHz, and 96 GB RAM. All models were implemented in PyTorch 1.11.0 and optimized using Adam. Following the BatteryLife setting (Tan et al., 2025), the model input was fixed to the first 100 cycles, and each cycle was represented by 300 sampled points after preprocessing.

Training was performed for at most 100 epochs with early stopping, and the patience was set to 5. Hyperparameters were selected on the validation split and then kept fixed during testing. For DITING, each cycle contains three variables, and the tri-coupled manifestation module uses a hidden size of 64. The health prototype is constructed from the first 8 early cycles, with 24 OT bins, an OT-softmin temperature of 0.15, 50 Sinkhorn iterations, and a cross-channel cost coefficient of 0.25. The Sinkhorn regularization coefficient is set to 0.08. The dynamical module is initialized with damping 0.35, coupling strength 0.75, drive amplitude 0.50, drive frequency 0.15, residual injection scale 1.00, step size 0.03, and state initialization scale 10^{-3} . These settings determine the prototype construction and the discrete-time dynamical update, while all learnable parameters are optimized jointly during training. We further provide a sensitivity analysis of four key hyperparameters, including prototype cycles, OT-softmin temperature, Sinkhorn iteration budget, and cross-channel penalty, with visualizations on some datasets shown in Figure 7.

D.5. Additional Results and Analysis

We provide additional analyses to complement the main experimental results. We first examine the representation geometry before and after tri-coupled manifestation, then report the reduced-cycle evaluation on datasets, and finally evaluate robustness under synthetic input noise. These results provide further evidence for the effectiveness of DITING under limited early-cycle observations and stochastic perturbations.

Representation geometry. To examine how tri-coupled manifestation reshapes degradation-related representations, Figure 8 compares the representation geometry before and after manifestation. In the first panel, the residual representations before

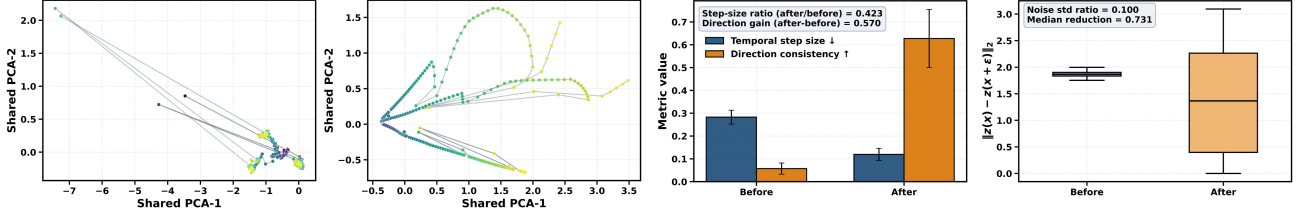


Figure 8. Representation geometry before and after tri-coupled manifestation. From left to right, the panels show the pre-manifestation representation geometry, the post-manifestation representation geometry, geometry statistics in the representation space, and noise sensitivity in the representation space.

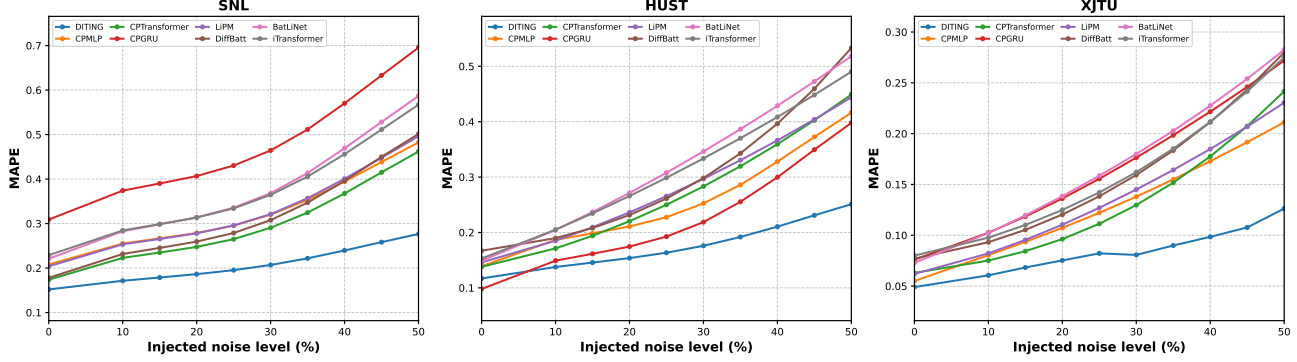


Figure 9. Evaluation under input noise. MAPE trends are reported on three datasets as the injected noise level increases from 0% to 50%.

manifestation are relatively scattered and contain larger local jumps, indicating that weak degradation cues are still mixed with stochastic disturbances. In the second panel, the trajectories after manifestation become smoother and more directionally organized in the shared PCA space, suggesting that the representations evolve along more coherent degradation-related paths. The third panel further quantifies this effect through two geometry statistics. After manifestation, the average temporal step size decreases, while directional consistency increases, showing that the module reduces erratic local variation while strengthening the stable cross-cycle direction. The fourth panel injects input noise and measures the representation perturbation $\|z(\mathbf{x}) - z(\mathbf{x} + \epsilon)\|_2$. The representations after manifestation show lower median sensitivity, indicating that the module suppresses inconsistent stochastic perturbations and improves representation robustness.

Prediction with Fewer Cycles. We select six datasets and provide the complete reduced-cycle evaluation results in Table 5. This setting examines whether DITING remains reliable when fewer early cycles are available. DITING remains competitive under shortened input windows, showing its ability to extract degradation evidence from limited observations.

Evaluation under Input Noise. To further examine tolerance to measurement disturbances, we inject synthetic stochastic noise into the 100-cycle inputs for three datasets at noise levels of 10%, 15%, 20%, 25%, 30%, 35%, 40%, 45%, and 50%. Figure 9 reports the MAPE trends as the injected noise level increases. Across the three datasets, DITING shows a more gradual performance degradation than the baselines, consistent with its discrepancy-aware design that emphasizes stable degradation evidence under stochastic perturbations.

E. Experiments in Other Domains

To further examine whether DITING can generalize beyond battery lifetime prediction, we conduct an additional validation on the C-MAPSS benchmark for aircraft engine remaining useful life prediction. C-MAPSS contains multivariate sensor measurements with operational variations and sensor noise, making it a representative testbed for weak signal extraction under noisy degradation processes. We evaluate DITING on the four standard subsets, FD001–FD004, and compare it with recent engine remaining useful life prediction baselines (Lin et al., 2024; Wang et al., 2024b; Fu et al., 2025). We report two standard metrics, RMSE and Score. RMSE measures the overall prediction error, while Score is an asymmetric prognostic metric that penalizes late and early predictions differently. Lower values indicate better performance for both metrics. As shown in Table 6, DITING achieves strong performance across all four subsets, indicating that the discrepancy-aware

Table 5. Fewer-cycle lifetime prediction across six datasets.

DATASET	CYCLES	DITING	CPMLP	CPTRANSFORMER	CPGRU	LiPM	DIFFBATT	BATLINET	ITRANSFORMER
CALCE	20	0.291	<u>0.350</u>	0.368	0.422	0.378	0.365	0.392	0.399
	40	0.263	<u>0.288</u>	0.327	0.359	0.322	0.308	0.329	0.342
	60	0.228	0.252	0.268	0.310	0.276	<u>0.248</u>	0.293	0.301
	80	0.204	0.237	0.238	0.276	0.263	<u>0.231</u>	0.252	0.274
	100	0.187	<u>0.221</u>	0.226	0.247	0.257	0.225	0.233	0.256
SNL	20	0.258	0.366	<u>0.312</u>	0.521	0.345	0.333	0.382	0.352
	40	0.206	0.287	<u>0.239</u>	0.418	0.267	0.259	0.297	0.276
	60	0.178	0.246	<u>0.203</u>	0.353	0.232	0.214	0.258	0.238
	80	0.159	0.224	<u>0.186</u>	0.332	0.214	0.189	0.241	0.232
	100	0.152	0.208	<u>0.174</u>	0.309	0.204	0.178	0.221	0.229
HNEI	20	0.060	0.073	0.071	0.086	<u>0.061</u>	0.078	0.084	0.088
	40	<u>0.047</u>	0.055	0.056	0.068	0.043	0.061	0.066	0.069
	60	0.040	0.046	0.047	0.057	<u>0.041</u>	0.060	0.055	0.057
	80	0.036	<u>0.040</u>	0.041	0.049	0.036	0.045	0.047	0.051
	100	0.035	<u>0.036</u>	<u>0.036</u>	0.037	0.035	0.037	0.038	0.038
HUST	20	<u>0.188</u>	0.220	0.217	0.159	0.239	0.243	0.248	0.260
	40	<u>0.152</u>	0.176	0.172	0.125	0.192	0.206	0.191	0.205
	60	<u>0.130</u>	0.151	0.146	0.112	0.167	0.191	0.166	0.174
	80	<u>0.121</u>	0.142	0.140	0.103	0.150	0.175	0.154	0.169
	100	<u>0.117</u>	0.139	0.138	0.098	0.146	0.167	0.149	0.153
XJTU	20	0.108	<u>0.132</u>	0.153	0.177	0.146	0.182	0.165	0.199
	40	0.084	<u>0.103</u>	0.119	0.135	0.111	0.146	0.130	0.157
	60	0.065	<u>0.078</u>	0.085	0.102	0.083	0.112	0.100	0.115
	80	0.058	<u>0.067</u>	0.072	0.084	0.075	0.089	0.082	0.092
	100	0.049	<u>0.055</u>	0.063	0.076	0.062	0.077	0.073	0.080
CALB	20	<u>0.298</u>	0.261	0.313	0.368	0.322	0.419	0.455	0.318
	40	<u>0.237</u>	0.208	0.241	0.288	0.252	0.340	0.396	0.253
	60	<u>0.198</u>	0.176	0.206	0.242	0.214	0.292	0.324	0.217
	80	<u>0.178</u>	0.155	0.186	0.212	0.195	0.270	0.303	0.190
	100	<u>0.169</u>	0.148	0.175	0.195	0.177	0.256	0.278	0.176

Table 6. Additional validation in other domains on the C-MAPSS dataset.

METHODS	FD001		FD002		FD003		FD004	
	RMSE	SCORE	RMSE	SCORE	RMSE	SCORE	RMSE	SCORE
C-T	12.80	234	17.61	1361	13.15	290	21.03	2303
DVGTFORMER	11.33	179	14.28	797	11.89	<u>254</u>	<u>14.50</u>	<u>894</u>
PSTFORMER	<u>11.08</u>	224	<u>12.25</u>	<u>575</u>	<u>11.34</u>	308	14.74	1182
DITING	10.56	<u>181</u>	11.26	498	10.23	195	12.36	629

mechanism can also benefit degradation prediction tasks beyond batteries. Since C-MAPSS involves different operating conditions, sensor configurations, and degradation regimes from battery datasets, the results provide additional evidence for the cross-domain applicability of the proposed manifestation mechanism. These findings further support the view that constructing discrepancy-aware representations is a general strategy for robust degradation prediction under noisy and weak-signal conditions.